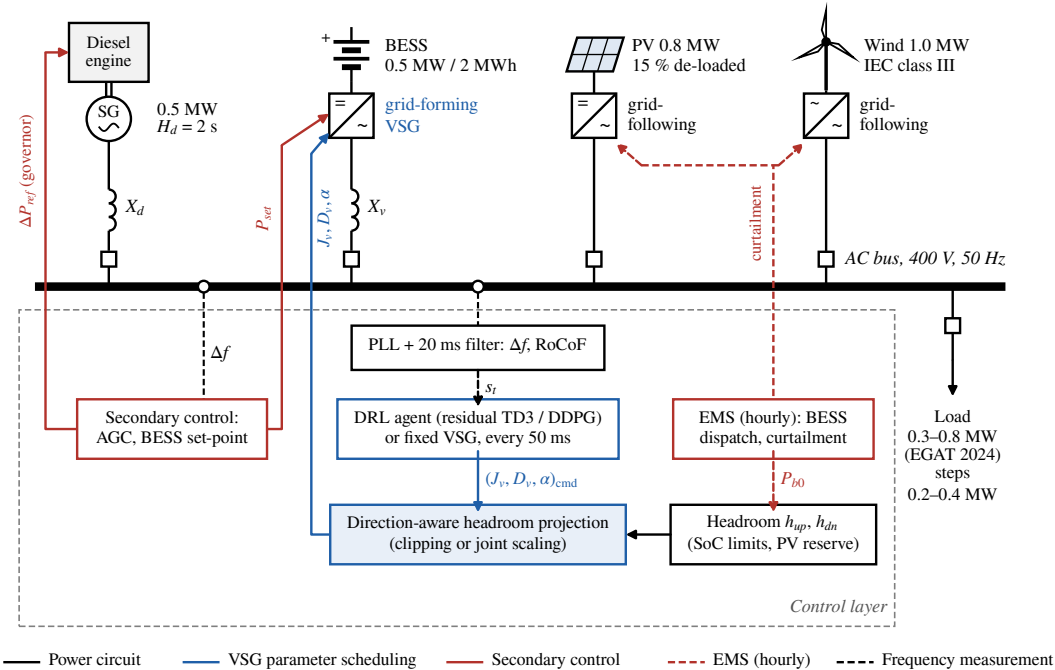
Diesel engine
SG
0.5 MW
H_d = 2 s
X_d
ΔP_{ref} (governor)
BESS
0.5 MW / 2 MWh
grid-forming
VSG
X_v
P_{set}
J_v, D_v, α
PV 0.8 MW
15 % de-loaded
grid-following
Wind 1.0 MW
IEC class III
grid-following
curtailment
AC bus, 400 V, 50 Hz
Δf
Secondary control:
AGC, BESS set-point
PLL + 20 ms filter: Δf, RoCoF
s_t
DRL agent (residual TD3 / DDPG)
or fixed VSG, every 50 ms
$(J_v, D_v, \alpha)_{cmd}$
Direction-aware headroom projection
(clipping or joint scaling)
EMS (hourly): BESS
dispatch, curtailment
P_{b0}
Headroom h_{up}, h_{dn}
(SoC limits, PV reserve)
Load
0.3–0.8 MW
(EGAT 2024)
steps
0.2–0.4 MW
Control layer
Power circuit
VSG parameter scheduling
Secondary control
EMS (hourly)
Frequency measurement